

Project Design Phase

Solution Architecture

Date	18 July 2026
Team ID	
Project Name	Nutrition Assistant
Maximum Marks	4 Marks

Solution Architecture

Solution architecture is a complex process — with many sub-processes — that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Nutrition Assistant – Solution Architecture Diagram (described)

Presentation Layer: React 19 + Vite client (Tailwind CSS, Framer Motion, Chart.js) served as a static site.

Application Layer: Express.js REST API behind Helmet, CORS and rate-limiting middleware; JWT authentication and Zod validation gate each protected route; MVC-layered controllers implement business logic (auth, tracker, recipes, meals, foods, settings, AI chat, admin).

Data Layer: MongoDB, accessed via Mongoose ODM, storing Users, Foods, Recipes, Meals, NutritionLogs, WaterLogs, Favorites and Settings collections, hosted on MongoDB Atlas.

External Services: Google Gemini API (gemini-1.5-flash) for the AI Nutrition Chatbot, with an intelligent rule-based fallback; Cloudinary for image storage, with local disk storage as a fallback via Multer.

Figure: Client (React + Vite) → HTTPS/REST (JSON) → Server (Node.js + Express — Auth Middleware, Role Guards, Controllers) → Mongoose ODM → MongoDB Atlas, with parallel calls out to the Gemini API and Cloudinary.

Reference: <https://aws.amazon.com/blogs/industries/> (solution architecture patterns)